

13 The density of helium at 150 kPa is 0.178 kg m^{-3} . What is the root-mean-square speed of its particles?

A 130 m s^{-1}

B 232 m s^{-1}

C 1300 m s^{-1}

D 1600 m s^{-1}

Ans: D

From the kinetic theory of gases,

$$p = \frac{1}{3} \frac{Nm}{V} \langle c^2 \rangle$$

$$p = \frac{1}{3} \rho \langle c^2 \rangle$$

$$c_{r.m.s.} = \sqrt{\frac{3p}{\rho}} = \sqrt{\frac{3(150 \times 10^3)}{0.178}} = 1600 \text{ m s}^{-1}$$

14 An ideal gas goes through the thermodynamic cycle EFGHE as shown